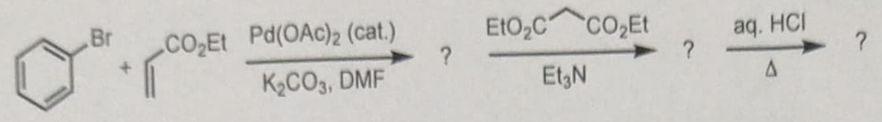
INDIAN INSTITUTE OF SCIENCE EDUCATION AND RESEARCH (IISER) MOHALI

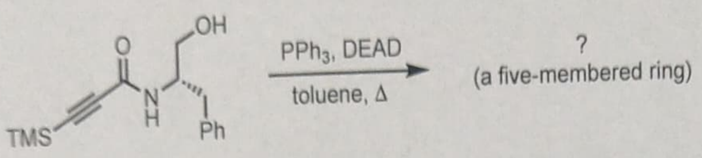
CHM302 Final Exam

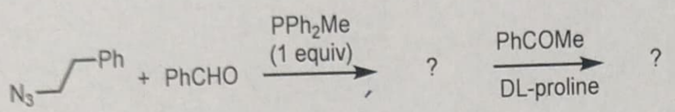
Total: 50 marks. Time: 3 Hours

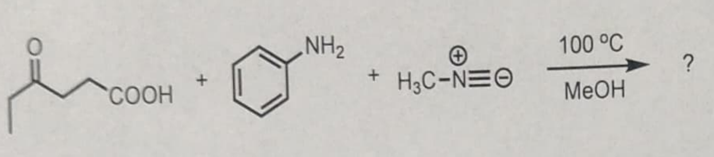
Answer all questions

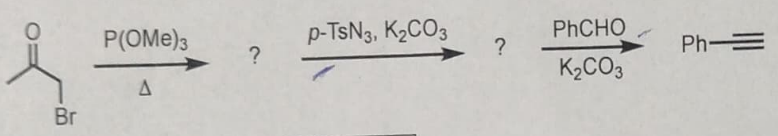
1. Write the expected product(s) in the following transformations with detailed mechanism at each step. (6 x 6 = 36)

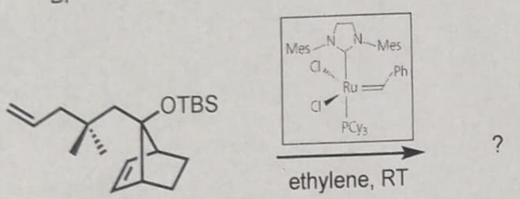
(a) 

(b) 

(c) 

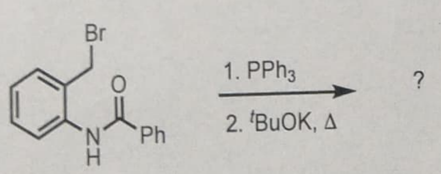
(d) 

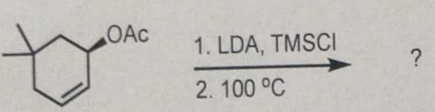
(e) 

(f) 

2. Write the products formed in the reactions of diazomethane with the following compounds, and provide detailed mechanisms for each reaction: (i) Cyclopentanone, (ii) Benzoic acid, and (iii) Benzoyl chloride (6)

3. Write the expected product(s) in the following transformations with detailed mechanism at each step. (2 x 4 = 8)

(g) 

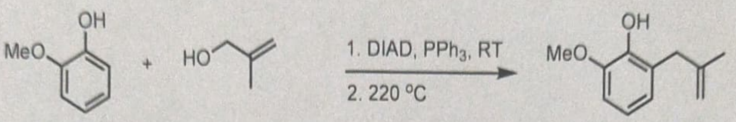
(h) 

INDIAN INSTITUTE OF SCIENCE EDUCATION AND RESEARCH
(IISER) MOHALI

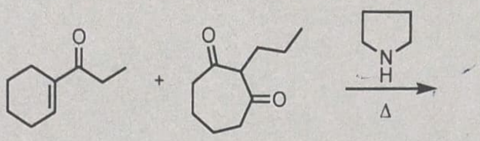
CHM302 2nd Mid-Sem Exam

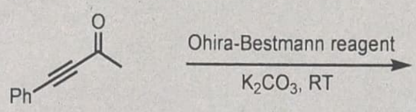
Answer all questions **Total: 20 marks. Time: 1 Hour**

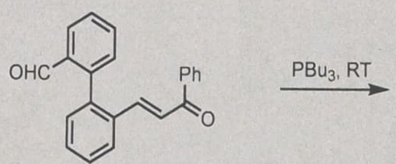
1. Write <u>detailed mechanism</u> of the following transformation. (4)



2. Predict the possible product(s) in the following transformations with <u>detailed mechanism</u> (3x4 = 12)







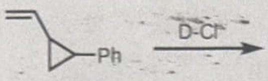
3. Suggest a method to synthesize 1,2-aminoalcohols and 3-amino-2-hydroxyesters (4)

INDIAN INSTITUTE OF SCIENCE EDUCATION AND RESEARCH (IISER) MOHALI

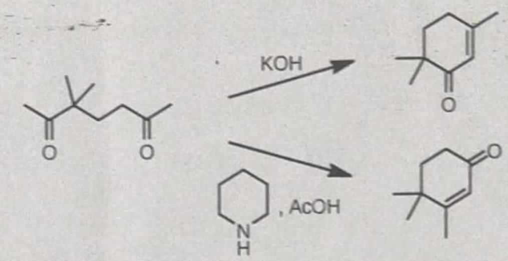
CHM302 1st Mid-Sem Exam

Answer all questions **Total: 20 marks. Time: 1 Hour**

1. Write all possible products in the following reaction by invoking a non-classical cationic intermediate. (4)

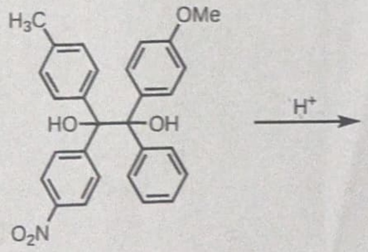


2. Write detailed mechanism of the following transformations. (6)

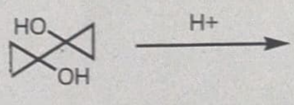


3. Write the expected product(s) in the following transformations with complete mechanistic details. (2 x 3 = 6)

(a)



(b)



4. Explain the origin of *syn*- and *anti*-aldol products from (*Z*) and (*E*)-enolates using Zimmerman–Traxler model. (4)